

Article

A Unified Experimental Database and Group-Aware Pilot Machine Learning for Flow Boiling in Rectangular Minichannels

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Abstract

A unified experimental database and a documented Python (3.12.6)-based harmonisation workflow are presented for local heat-transfer analysis in rectangular-minichannel flow boiling. The database integrates 449 experimental source files and 64,385,791 point-level records covering six working fluids, multiple surface conditions, channel configurations, and orientations. The workflow combines template-based files, central-line infrared wall-temperature data, geometric and operating metadata, local pressure and saturation temperature reconstruction, bulk fluid temperature interpolation, heat-loss correction, local heat-transfer coefficient and Nusselt number calculation, operational branch labelling, and auditable quality-control flags. The database is characterised at the point and source-file levels to identify branch imbalance, unequal source-file sizes, and heterogeneous coverage. A source-file-grouped pilot benchmark is conducted on streaming-sampled subcooled and saturated subsets using Random Forest regressors and five group-based train/test partitions. Across the five splits, the nine-predictor RF achieved $R^2 = 0.887 \pm 0.015$ for subcooled Nu and $R^2 = 0.810 \pm 0.043$ for the subcooled heat-transfer coefficient; saturated performance was weaker and more split-sensitive ($R^2 = 0.340 \pm 0.134$ for Nu and 0.317 ± 0.091 for the heat-transfer coefficient). These results are treated as feasibility screening rather than final model ranking or evidence of campaign- or configuration-independent transfer. The database architecture and group-aware pilot validation provide a documented foundation for controlled physical interpretation and further validation.

Keywords: flow boiling; rectangular minichannel; experimental database; data harmonisation; quality control; Nusselt number; heat transfer coefficient; source-file-level validation; pilot machine learning; Random Forest



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1. Introduction

Flow boiling in minichannel systems is an effective approach to compact thermal management because sensible and latent heat transport can provide high local heat-transfer rates within small flow passages. The measured response, however, depends simultaneously on working fluid properties, channel geometry, heated surface condition, spatial orientation, mass-flow level, pressure, heat flux, wall temperature, and the local boiling state. Experimental programmes conducted over several years therefore produce archives that are heterogeneous not only in physical coverage but also in file structure, metadata conventions, spatial resolution, and data reduction history.

For consistency of terminology, channels with characteristic dimensions of 200 μm –3 mm are classified here as minichannels, whereas channels in the 10–200 μm range are classified as microchannels. All rectangular channels included in the present archive fall within the minichannel range, with hydraulic diameter below 3 mm.

This heterogeneity becomes a methodological problem before any physical ranking or machine-learning model is attempted. Local records extracted from the same source file are strongly related, different experimental campaigns may contribute very different numbers of files and rows, and apparently large point-level sample sizes can therefore be dominated by a comparatively small number of provenance groups. A credible foundation for statistical analysis and machine learning requires explicit provenance, documented and traceable harmonisation, operational-branch-aware local reconstruction, auditable quality control, source-file-level diagnostics, and validation that prevents records from the same source file from appearing on both sides of the train/test boundary.

Consolidated Flow-Boiling Databases and Machine-Learning Methodology

Machine learning has developed rapidly in boiling heat-transfer research, including prediction of heat-transfer coefficients, critical heat flux, pressure drop, and boiling-regime characteristics. The recent review by Chu et al. [1] shows that this growth has been accompanied by recurring limitations: incomplete or unevenly distributed datasets, inconsistent source quality, overfitting, and weak extrapolation beyond the range represented during training. These limitations are especially important for flow-boiling archives because the target quantities are commonly reconstructed from measured and derived variables rather than observed directly. Previous work on related rectangular-minichannel systems also demonstrated that uncertainty in heated-wall temperature measurements should be propagated into the calculated heat-transfer quantities rather than considered independently of the reconstruction procedure [2].

A major advance was the construction of consolidated multi-source databases. Qiu et al. [3] trained an artificial neural network on 16,953 saturated flow-boiling records collected from 50 sources and 16 fluids. Their additional tests with complete databases withheld from training showed that predictions remained comparatively strong when the working fluid was represented elsewhere in the training data but deteriorated for a previously unseen fluid. Bard et al. [4] subsequently examined the same consolidated database using exploratory data analysis, missing-data imputation, multiple feature-selection procedures, and seven model families. Support vector regression produced the strongest overall result, while the analysis also exposed the influence of high-leverage experiments and extreme observations. Qiu et al. [5] then combined physically motivated candidate variables with Pearson correlation and mutual information to optimise an ANN input set, demonstrating that increasing the number of features does not improve performance monotonically.

Recent studies have expanded both the range of operating conditions and the model families considered. Mudawar et al. [6] used 29,226 heat-transfer records spanning highly subcooled to high-quality saturated flow boiling under microgravity and different Earth-gravity orientations, obtaining an overall ANN error of 7.99% and assessing whether the predicted trends remained physically plausible. Bediako and Elbarghthi [7] compared nine machine-learning models using dimensional and dimensionless representations for high-saturation-temperature flow boiling. Noh et al. [8] applied XGBoost to 11,470 pre-dryout records from 41 sources and 23 fluids, combining dryout filtering, permutation importance, SHAP analysis, and hyperparameter optimisation. Ayegba et al. [9] recently assembled 20,891 saturated-flow-boiling measurements from 83 studies and used physics-aligned features with neural-network, support-vector, random-forest, and gradient-boosting models. Their study achieved strong pooled performance but did not establish transfer to

unseen studies or fluids; the authors explicitly identified study-wise cross-validation and fluid-held-out tests as priorities for future work.

The validation design is therefore as important as the selected algorithm. Loyola-Fuentes et al. [10] proposed a six-stage framework for heat-transfer regression that explicitly includes data pretreatment, feature selection, splitting philosophy, model training, hyperparameter tuning, and performance assessment. In their case study, machine learning improved interpolation accuracy but extrapolated less reliably than a semi-empirical correlation. Mehdi et al. [11] reached a similar methodological conclusion for enhanced pool boiling: models that performed well on a conventional train/test split showed distinctly larger errors on six deliberately excluded fluid-surface datasets. Wang and Kharangate [12] further demonstrated that database size alone is not a measure of information content. Their pointwise uncertainty analysis distinguished epistemic uncertainty caused by insufficient coverage from aleatoric uncertainty associated with noise and showed that adding observations near high-uncertainty regions improved prediction much more than adding data in already dense regions.

Taken together, these studies identify four requirements for credible machine-learning analysis of heterogeneous boiling data: the source and composition of the database must remain traceable; regime filtering and feature construction must be physically justified; validation must reflect the intended level of transfer rather than merely interpolate between related records; and database coverage and uncertainty must be examined alongside aggregate error metrics. The literature also shows that strong pooled performance may coexist with weak transfer to a previously unseen fluid, surface, operating range, or experimental source.

The methodological progression is therefore moving from model-centred performance comparisons toward validation-aware and uncertainty-aware analysis. Nevertheless, most published workflows begin after a consolidated dataset has already been assembled, and several of the strongest reported results still rely on random record-level splitting. Such designs quantify interpolation within the pooled database but do not fully test transfer to an unseen experiment when many neighbouring local records share the same geometry, operating sequence, measurement chain, and data-reduction assumptions.

The remaining gap concerns the complete provenance-preserving path from heterogeneous experimental files to a modelling-ready local database. This path must harmonise evolving templates and metadata, preserve source-file identity, restructure spatial thermographic data, reconstruct local pressure and thermal reference quantities, define operational-branch-dependent heat-transfer outputs, retain explicit QC decisions, and characterise both point-level scale and source-file-level coverage before machine learning is applied.

The present article addresses this gap by documenting experimental provenance, the integrated database, the Python-based harmonisation and local reconstruction workflow, system-level quality control, global database diagnostics, and a source-file-grouped pilot Random Forest benchmark [13]. The present analysis is limited to the evidence required to evaluate the database architecture, data integrity, and first-stage predictive feasibility; it does not claim a definitive physical ranking of surfaces, orientations, fluids, or modelling approaches.

Accordingly, the objectives are: (i) to document the experimental archive and source-file provenance; (ii) to define the unified data model and the sequence of local thermal-hydraulic reconstruction; (iii) to describe the operational reference-temperature branches and auditable QC architecture; (iv) to quantify the composition, imbalance, and coverage of the integrated database at point and source-file levels; and (v) to establish pilot machine-learning baselines using source-file-level group separation rather than random row splitting.

2. Materials and Methods

2.1. Experimental Campaigns and Database Provenance

2.1.1. Experimental Facility and Representative Rectangular-Minichannel Module

The experimental data were obtained using a closed-loop flow-boiling facility comprising the working fluid circuit, the electrical power-supply and control systems, and synchronised thermal and visual measurement systems. The working fluid circuit included a gear pump controlled by a variable-frequency drive, a pressure-stabilisation tank, and a Coriolis mass flowmeter. The principal instrumentation also included inlet and outlet pressure transducers and temperature sensors. An infrared camera was used to observe the external surface of the heated plate, whereas a high-speed camera was positioned on the opposite side of the test section to record the flow through the glass plate.

Because the integrated database combines several experimental campaigns and test-section variants, no single drawing can represent every geometric configuration used in the experiments. Figure 1 therefore presents a representative rectangular-minichannel test section, identifies its principal structural components, and illustrates the channel-count configurations represented in the analysis. The experimental concept and infrared wall-temperature measurements build on the authors' earlier rectangular-minichannel investigations, including uncertainty-aware heat-transfer calculations [2] and comparative analyses based on Trefftz functions and Simcenter STAR-CCM+ [14].

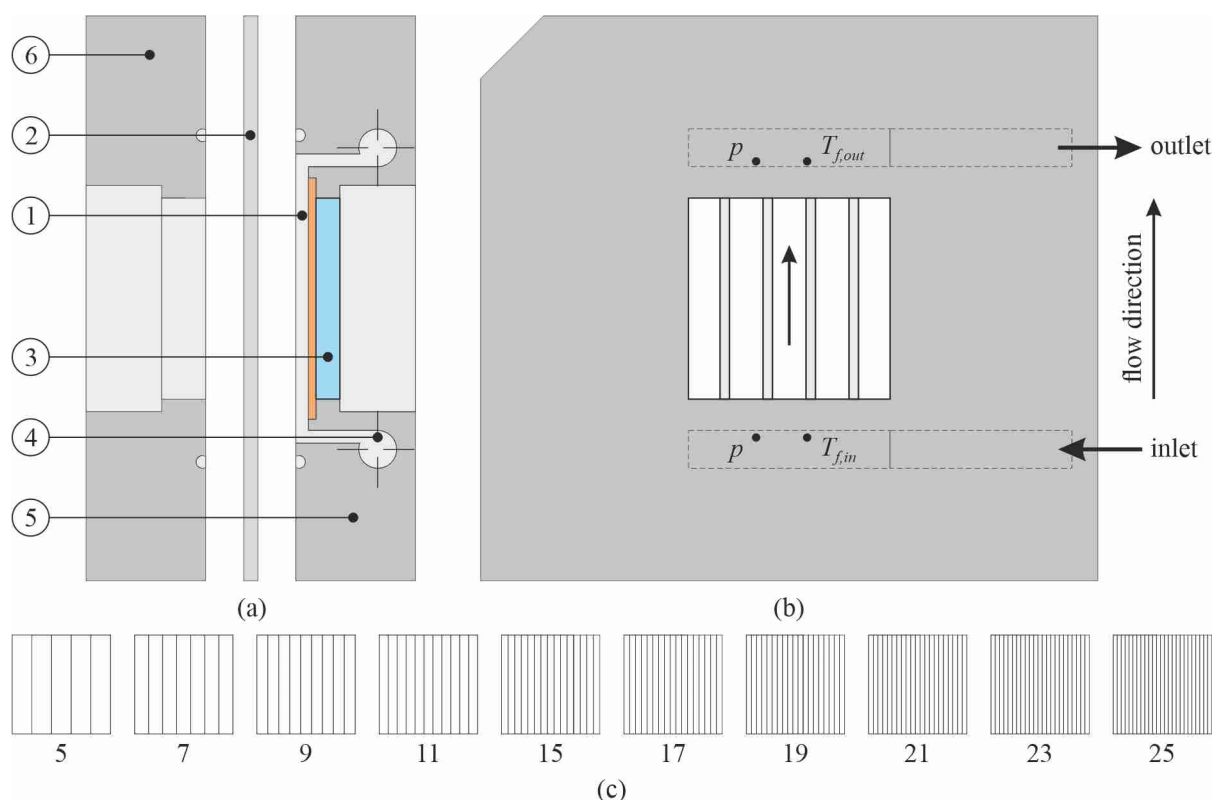


Figure 1. Representative rectangular-minichannel test sections: (a) longitudinal section showing the main components: 1—minichannel, 2—heated plate, 3—glass plate, 4—inlet manifold, 5—base, and 6—cover; (b) view from the cover side for the configuration with five minichannels; (c) channel-count configurations containing 5, 7, 9, 11, 15, 17, 19, 21, 23 and 25 parallel minichannels.

As shown in Figure 1, the test section had a modular structure that allowed the number of parallel minichannels to be varied while retaining the same general arrangement of the heated plate, glass plate, base, cover, and inlet manifold. The configurations presented in Figure 1c illustrate the geometric diversity represented in the integrated database and

should therefore be interpreted as selected module variants rather than as a single test-section geometry used throughout all experimental campaigns.

A recent direct comparison of liquid crystal thermography and infrared thermography in flow-boiling minichannels showed convergent heated-wall temperature distributions and average relative differences below 15% in the resulting local heat-transfer coefficients. This finding provides direct methodological support for the use of non-contact thermographic wall-temperature measurements as the thermal input to the present database workflow [15]. The relationship between the thermographic measurements and the principal hydraulic, thermal, and electrical measurements is illustrated schematically in Figure 2.

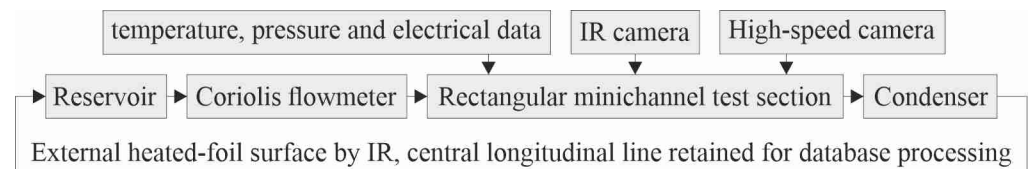


Figure 2. Simplified scheme of the closed-loop experimental facility and the associated thermal, hydraulic, electrical, and visual measurement chain.

Figure 2 shows that the infrared wall-temperature measurement formed only one component of the complete experimental measurement chain. The central longitudinal temperature profile extracted from the externally observed heated surface was linked to the corresponding source-file metadata, electrical power input, mass-flow rate, inlet and outlet fluid temperatures, pressure measurements, working fluid identity, surface condition, test-section configuration, and spatial orientation. The diagram is intentionally simplified and presents the general measurement and data-flow architecture rather than the complete arrangement of every facility component or a single geometric configuration common to all 449 source files.

2.1.2. Scope and Heterogeneity of the Experimental Archive

The integrated archive contains 449 processed experimental source files. The archive covers the following six dielectric working fluids: FC-72, FC-770, HFE-649, HFE-7000, HFE-7100, and HFE-7200, together with multiple surface identifiers, different channel-count configurations, and a broad range of nominal module orientations. Table 1 summarises the high-level experimental and database scope that defines the input domain of the foundation study.

Terminology used in this article reflects the hierarchical structure of the archive. A point-level record represents one local streamwise observation derived from the central infrared-temperature line and linked to the corresponding operating data. A thermal frame denotes one thermographic acquisition, while an operating state denotes the associated set of recorded boundary conditions used in the local reconstruction. A source file is the processing and provenance unit retained by the workflow. A campaign represents a broader measurement programme that may generate multiple source files, whereas an experiment or run denotes an individual physical measurement sequence documented in the available metadata. These levels are not equivalent; in particular, a source file is not assumed to represent an independent experimental campaign.

Table 1 shows that the database represents a heterogeneous experimental archive rather than a single-condition benchmark. Because fluids, surfaces, orientations, and configurations are not represented in a balanced way, pooled averages should not be interpreted as direct causal effects of these factors. The present study therefore uses the archive primarily to characterise data coverage, provenance, quality-control behaviour, and the scope and limitations of the group-aware pilot modelling.

Table 1. High-level scope and heterogeneity of the integrated experimental archive.

Database Descriptor	Current Value/Scope	Role in the Present Article
Processed experimental source files	449	Primary provenance groups
Point-level records	64,385,791 in the final integrated database	Local reconstructed records
Working fluids	6	Coverage and stratification variable
Working fluids included	FC-72, FC-770, HFE-649, HFE-7000, HFE-7100, HFE-7200	Fluid identity
Surface conditions	Smooth and multiple modified/developed identifiers	Database heterogeneity; no pooled physical ranking here
Nominal orientations	Multiple horizontal, vertical and inclined positions	Coverage variable
Channel-count configurations	Multiple configurations	Geometry metadata
Primary local outputs	α and Nu	Targets for diagnostics and pilot ML

To make the heterogeneity visually explicit, Figure 3 illustrates the experimental archive from source files to the principal experimental descriptors and downstream data products. The figure is intended to show how a moderate number of provenance groups expands into a very large point-level database after spatial restructuring and local reconstruction.

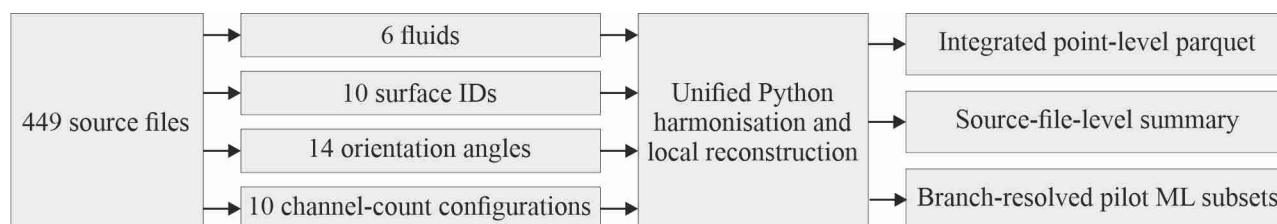
**Figure 3.** Data provenance and database composition overview from 449 source files to local reconstructed records and analysis subsets.

Figure 3 emphasises two methodological facts. First, all local records remain traceable to their source file, which is used as the grouping unit in the pilot validation. Second, the point-level scale of the database is generated by repeated local observations nested within a much smaller number of source files, not by millions of independent experiments. Source-file separation prevents records from the same file from occurring on both sides of the train/test boundary; it does not, by itself, guarantee independence between related campaigns, test-section configurations, or operating sequences.

2.2. Unified Data Model and Python-Based Harmonisation Workflow

2.2.1. Source Files, Harmonisation, and Output Architecture

The experimental source files were processed through a modular Python workflow designed for large batch runs and repeatable reprocessing. The workflow reads template-based experimental files, extracts geometry and operating metadata, restructures the central longitudinal infrared temperature line into local records, merges the thermal profile with frame- or state-level operating data, reconstructs local thermodynamic quantities, applies operational-branch and quality-control logic, and writes columnar outputs suitable for subsequent analysis.

The complete processing sequence is summarised in Figure 4. The purpose of the figure is to separate the data-engineering stages from the later physical and machine-learning analyses and to make the lineage of each derived variable explicit.

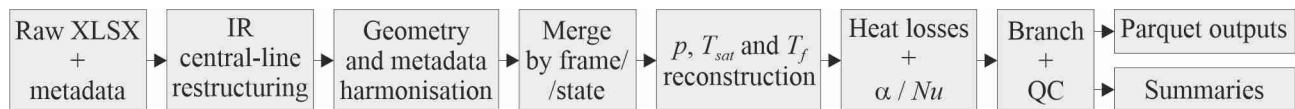


Figure 4. Unified Python-based data-processing workflow from raw experimental files to the integrated Parquet database, QC outputs, source-file-level summaries, and pilot ML subsets.

As Figure 4 shows, the database is not produced by a single transformation but by a sequence of traceable operations. The separation of raw inputs, intermediate thermodynamic reconstruction, derived heat-transfer quantities, QC flags, and final analytical subsets allows the pipeline to be rerun when a definition or threshold is revised. The use of a columnar Parquet output also permits selective reading and batch analysis without loading the complete dataset into memory.

2.2.2. Unified Variable Groups

For traceability, the merged dataset is organised into clearly defined variable groups rather than treated as an undifferentiated table. Table 2 summarises these groups, provides representative variables, and specifies their analytical roles. This structure also supports subsequent feature auditing by clearly distinguishing measured inputs, reconstructed thermodynamic quantities, derived heat-transfer outputs, provenance identifiers, and quality-control variables.

Table 2. Variable groups in the unified point-level database and their analytical roles.

Variable Group	Representative Variables	Primary Purpose
Provenance	Source file and experiment identifiers	Data grouping, traceability, and prevention of source-file leakage
Geometry	Channel dimensions, hydraulic diameter, observation length, and number of minichannels	Local thermal-hydraulic reconstruction and definition of geometric similarity variables
Operating conditions	Mass flow rate, electric current and voltage, working fluid inlet and outlet temperatures, inlet and outlet gauge pressures, and atmospheric pressure	Definition of hydraulic, thermal, and electrical boundary conditions
Thermographic line data	Central-line wall temperature and streamwise coordinate x	Spatially resolved thermal input
Reconstructed thermal/reference quantities	Local absolute pressure, corresponding local saturation temperature of the working fluid, interpolated bulk-fluid temperature, and selected reference temperature	Reconstruction of local pressure, saturation temperature and branch-dependent reference temperature
Heat-flux terms	Electrical heat input, wall heat flux, net heat flux, and related heat-flux descriptors	Energy-balance calculations and construction of heat-transfer outputs
Derived heat-transfer quantities	Local heat-transfer coefficient α , Nusselt number Nu , Reynolds number Re , Prandtl number Pr , and related descriptors	Analysis and pilot ML
Operational branch and QC	Operational branch label, QC_{flags} , QC_{keep}	Operational-branch-specific filtering, quality assessment, and analytical-subset definition

Table 2 makes clear that the local heat-transfer coefficient α and Nusselt number Nu are derived quantities rather than direct sensor measurements. This distinction is important in the pilot machine-learning analysis because some candidate input variables, particularly heat-flux descriptors, are mathematically related to the target definitions and should therefore be treated as target-proximal features rather than as independent physical predictors. The database and code identifiers used in Figure 5, including T_{wall} , $T_{f,used}$, q_w , and q_{net} , are mapped to the corresponding scientific symbols listed in the Nomenclature.

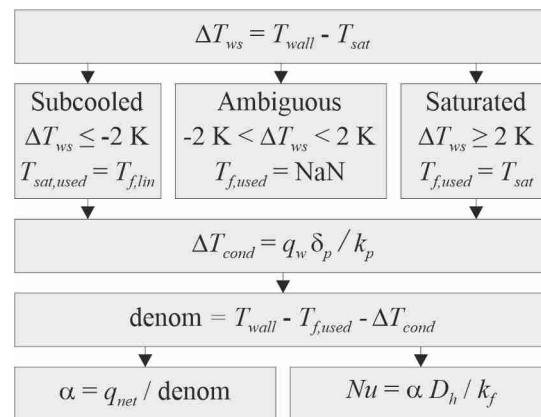


Figure 5. Operational-branch-aware reconstruction of local thermal quantities.

2.2.3. Local Thermal Reconstruction and Operational Reference-Temperature Branches

The local heat-transfer calculation requires a reference fluid temperature that changes with the local thermal state. The implemented workflow therefore separates the subcooled and saturated branches rather than imposing one denominator over the complete database. Figure 5 summarises the two branches and identifies the variables entering the local heat-transfer coefficient and Nusselt number calculations. Because α and Nu depend on reconstructed temperature differences, uncertainty in the heated-wall and fluid-temperature measurements propagates directly into these derived quantities and becomes especially influential when the effective denominator is small. In a related rectangular-minichannel study, temperature-measurement uncertainty was evaluated using both Monte Carlo simulation and the uncertainty-propagation method and was incorporated into the heat-transfer coefficient calculations [2].

The recorded inlet and outlet pressures are gauge pressures. Absolute inlet and outlet pressures are obtained by adding atmospheric pressure (i.e., $p_{abs,in} = p_{g,in} + p_{atm}$, and $p_{abs,out} = p_{g,out} + p_{atm}$).

Local absolute pressure is reconstructed by linear interpolation between the inlet and outlet values along the streamwise channel coordinate x , where L is the heated channel length:

$$p_{abs}(x) = p_{abs,in} + (x/L)[p_{abs,out} - p_{abs,in}] \quad (1)$$

The reconstructed local absolute pressure $p_{abs}(x)$ is subsequently used to determine the local saturation temperature $T_{sat}(x)$. In fluids.py, this calculation uses the fluid-specific producer vapour-pressure relations for FC-72, FC-770, Novec 649, Novec 7000, Novec 7100, and Novec 7200. For each fluid, the corresponding analytical manufacturer correlation is inverted at $p_{abs}(x)$ to obtain $T_{sat}(x)$. The property provenance is retained explicitly in the workflow as producer data [16–21].

The local bulk-fluid reference temperature is reconstructed by linear interpolation between the measured inlet and outlet fluid temperatures:

$$T_{f,lin}(x) = T_{f,in} + (x/L)[T_{f,out} - T_{f,in}] \quad (2)$$

$T_{f,lin}(x)$ is therefore a reconstructed local bulk-fluid reference, not a directly measured local fluid temperature, and is used as the reference temperature for the operational subcooled branch.

The implemented Python module (`alpha_nu.py`) evaluates the wall-to-saturation temperature difference and assigns the operational analytical branches according to the adopted criteria: (i) $\Delta T_{ws} = T_{wall} - T_{sat}$; (ii) $\Delta T_{ws} \leq -2$ K (operational subcooled branch); and (iii) $\Delta T_{ws} \geq +2$ K (operational saturated branch) and $|\Delta T_{ws}| < 2$ K (ambiguous branch).

For records assigned to the ambiguous branch, $T_{f,used}$ is set to NaN and α and Nu are not calculated. The selected reference temperature is $T_{f,lin}$ for the operational subcooled branch and T_{sat} for the operational saturated branch.

The one-dimensional heated-wall conduction correction applied in the established data reduction workflow is:

$$\Delta T_{cond} = q_w \cdot \delta_p / k_p \quad (3)$$

The heated wall is a HAYNES 230 alloy foil. The historical data-reduction workflow used a constant heated-plate thermal conductivity of $k_p = 8.3 \text{ W m}^{-1} \text{ K}^{-1}$, as implemented in `alpha_nu.py`. Manufacturer data indicate that the thermal conductivity of HAYNES 230 varies with temperature [22]. This constant value is retained in the present article to preserve consistency across the integrated archive and is treated as a simplifying reconstruction assumption. The one-dimensional conduction correction should therefore be interpreted as a standardised data-reduction step rather than as a complete representation of multidimensional heat conduction in every experimental configuration.

The database variable q_{net} denotes the net heat flux transferred to the fluid after applying the standardised physics-based external heat-loss model to the wall/electrical heat-flux term. The external heat loss flux is evaluated record by record as the sum of natural-convection and radiation contributions:

$$q_{loss,ext} = q_{conv,ext} + q_{rad,ext} \quad (4)$$

The natural-convection contribution is evaluated using the Churchill–Chu vertical-plate relation [23], with air properties evaluated at the film temperature:

$$T_{film} = \frac{(T_{wall} + T_{amb})}{2} \quad (5)$$

The radiative contribution is calculated as:

$$q_{rad,ext} = \varepsilon \cdot \sigma \cdot [T_{wall}^4 - T_{amb}^4] \quad (6)$$

assuming $\varepsilon = 0.97$.

The external-loss flux is converted to the heated-area reference using the ratio of the external area to the heated area:

$$q_{loss} = q_{loss,ext} \cdot \left(\frac{A_{ext}}{A_{heated}} \right) \quad (7)$$

where $A_{ext} = L_{obs} w$.

The net heat flux used in the heat-transfer reconstruction is then:

$$q_{net} = q_w - q_{loss} \quad (8)$$

The same standardised loss model is applied consistently across the integrated experimental series. It should therefore be interpreted as a standardised archive-wide correction model rather than as an orientation-specific external-convection model. Because the loss is

calculated record by record from wall temperature, ambient temperature, and geometry, no single constant heat-loss magnitude applies to the complete archive. The workflow did not materialise a separate archive-wide uncertainty value for this correction.

The effective temperature-difference denominator and the derived heat-transfer quantities are defined as:

$$\text{denom} = T_{\text{wall}} - T_{f,\text{used}} - \Delta T_{\text{cond}} \quad (9)$$

$$\alpha = \frac{q_{\text{net}}}{\text{denom}} \quad (10)$$

$$Nu = \frac{\alpha \cdot D_h}{k_f} \quad (11)$$

These categories define operational data-reduction branches and should not be interpreted as a complete classification of the bulk thermodynamic state or flow pattern.

2.2.4. Operational Branch Labelling and Analytical Use

The operational branch label stored for each local record is a data-reduction variable used to select the reference-fluid temperature and to construct branch-specific analytical and machine-learning subsets. Records within the ambiguous deadband remain in the integrated archive for traceability but are excluded from the calculation of α and Nu and from branch-specific modelling. The labels' operational subcooled branch, operational saturated branch, and ambiguous branch are not flow-pattern labels and do not constitute a complete classification of the local or bulk thermodynamic state.

Positive saturated equilibrium vapour quality was not used as a predictor, was not materialised or released for the present analysis, and was not used for operational branch assignment. Accordingly, the operational saturated branch is defined by the adopted wall-to-saturation-temperature criterion rather than by a positive vapour-quality threshold. The absence of positive saturated vapour quality is treated as a limitation of the present saturated predictor set; it is not retrospectively reconstructed in the present analysis.

2.3. Quality-Control Architecture

QC Logic and Record-Level Flags

Quality control is implemented as a set of explicit flags rather than as an irreversible deletion step. Records may therefore remain present in the full merged database while being excluded from a specific analytical subset. Figure 6 presents the intended decision structure from operational-branch assignment through QC checks to the final record-retention status.

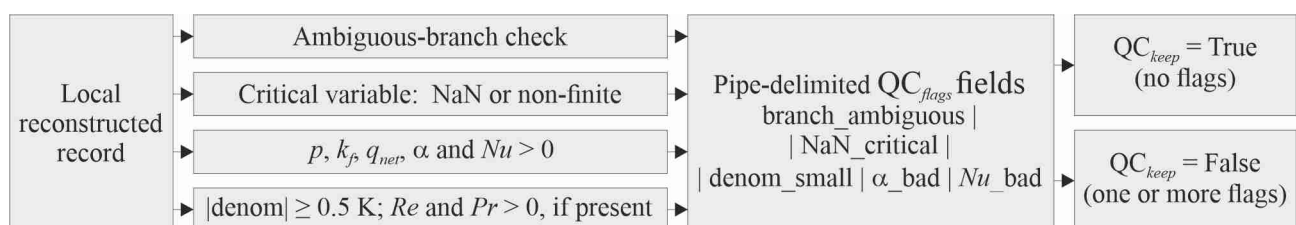


Figure 6. Operational-branch labelling and quality-control framework from the locally reconstructed record to the final record-retention status.

The ± 2 K deadband and the 0.5 K denominator limit were adopted as operational thresholds within the data-reduction and quality-control workflow. The former separates the subcooled and saturated branches from the ambiguous region around the saturation condition, whereas the latter prevents the calculation of α and Nu when the effective temperature-difference denominator approaches zero. These thresholds should be in-

terpreted as methodological settings of the present workflow rather than as universal physical limits.

Figure 6 emphasises that quality control is not equivalent to generic outlier removal. Some records are excluded by methodological design, for example, when operational-branch assignment remains ambiguous and no valid thermal reference temperature can be assigned, whereas other exclusions result from missing, non-finite, or physically invalid reconstructed quantities. Exclusion due to methodological ambiguity does not necessarily indicate an experimental or measurement failure.

The implemented QC rules are summarised in Table 3 and illustrated in Figure 6. The Python module (qc.py) constructs a pipe-delimited QC_{flags} field and assigns $QC_{keep} = \text{True}$ only when no quality-control flag is present. The default QC gate excludes records with an ambiguous operational-branch assignment; missing or non-finite critical variables; non-positive local absolute pressure, fluid thermal conductivity, net heat flux, heat-transfer coefficient α , or Nusselt number Nu ; an absolute effective temperature-difference denominator below 0.5 K; and non-positive Reynolds or Prandtl numbers when these quantities are available.

Table 3. Quality-control categories, record-level actions, and interpretation.

QC Category	Typical Condition	Effect on Record	Interpretive Meaning
Operational-branch ambiguity	Reference-temperature branch not uniquely assigned	Flagged and excluded from the retained and operational-branch-specific α and Nu subsets; preserved in the full archive	Methodological ambiguity, not necessarily a measurement error
Missing/invalid input	Required input unavailable or non-finite	Flagged and excluded from the retained subset	Insufficient data
Invalid reconstruction	Non-finite or unstable derived denominator/quantity	Flagged and excluded	Derived quantity not considered reliable
Physical plausibility	Value outside accepted physical/processing limits	Flagged and excluded from the retained subset; preserved in the full archive for review	Potential measurement or reconstruction problem
Valid retained record	Required inputs and derived outputs pass all checks	$QC_{keep} = \text{True}$	Eligible for global diagnostics and predefined analytical subsets

Table 3 distinguishes the reasons for record exclusion from the final retention status. This distinction allows the system-level QC decisions to remain auditable without altering the underlying data provenance or raw measurements. Detailed classification of infrared line-profile anomalies is outside the scope of this database-level QC framework and is not required for interpreting the record-retention logic reported here.

2.4. Source-File-Grouped Pilot Machine-Learning Methodology

Pilot Objective and Sampling Strategy

The machine-learning component of the present study is intentionally limited to a pilot screening task. Its purpose is to test whether the harmonised database contains learnable signal under source-file-grouped validation, not to establish a final ranking of algorithms or universal transferability. Because the complete merged dataset is too large for direct in-memory tree training, the pilot data were constructed by streaming the Parquet outputs and sampling within source-file groups.

The validation principle is illustrated in Figure 7. All records originating from a given source file are assigned to one side of the train/test boundary, so that closely related local points from that file cannot appear in both sets.

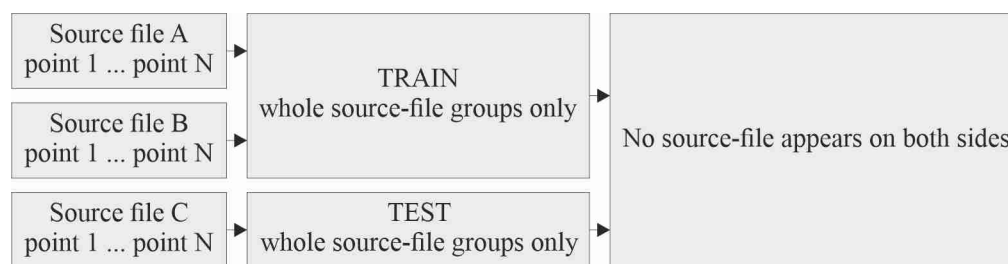


Figure 7. Source-file-separated train/test design for local point-level machine learning.

Figure 7 shows that the grouping unit used for validation is the experimental source file, even though the regressor is trained on point-level rows. This is more conservative than random row splitting because all records from the same source file remain in one partition. However, files produced within a shared campaign or configuration may still be related; consequently, the design controls direct within-file record leakage but should not be interpreted as a complete campaign- or configuration-independent transfer test.

The complete database comprised 449 source files. The pilot datasets were constructed by group-balanced streaming sampling directly from the Parquet outputs. Fifty valid records were sampled from each source file containing eligible records for the respective operational branch. This procedure yielded 22,350 subcooled records from 447 source files and 22,050 saturated records from 441 source files; the remaining files did not contain valid records for the corresponding branch after operational-branch and QC filtering. All sampled records originating from a given source file were retained within a single data partition. Source-file groups, rather than individual point-level rows, were assigned to the training or test subset, thereby preventing records from the same source file from occurring on both sides of the validation boundary.

Five source-file-grouped train/test partitions were generated with GroupShuffleSplit (test fraction 0.20) using split seeds 42, 43, 44, 45, and 46. For every split, complete source-file groups, rather than point-level rows, were assigned to training or test data. The same five source-file partitions were used for all Random Forest feature-set comparisons and the corresponding literature-reference evaluations. All four RF tasks used the same nine predictors as follows: Re , Pr , q_{net} , $T_{f,used}$, $T_{f,lin}$, T_{sat} , D_h , L_{obs} , and θ . The predictor q_w was excluded from every task. The field source_file was used exclusively as the grouping variable and was not supplied to the regressor. Performance is reported as the mean $\pm$ standard deviation across the five splits.

The pilot Random Forest implementation used 20 trees ($n_{estimators} = 20$), a maximum depth of 8, a minimum of 10 samples per leaf, square-root feature sampling at each split, bootstrap resampling with $max_samples = 0.3$, and $random_state = 42$. These deliberately conservative settings were adopted for the feasibility-screening stage and to permit memory-safe processing of the streamed samples. No hyperparameter optimisation was performed in this pilot analysis.

The nine-predictor set remains target-proximal because q_{net} and reconstructed thermal-state descriptors contribute directly or closely to the target definitions. It is therefore compared with the pre-specified strict baseline rather than interpreted as an independent physical predictor set.

A strict predictor baseline containing only Re , Pr , L_{obs} , and θ was evaluated with the same Random Forest configuration and the same five source-file-grouped partitions.

As supporting literature benchmarks, Sieder–Tate [24] was used for the subcooled branch as a laminar, thermally developing single-phase reference, rather than as a physical model of the entire subcooled-boiling branch. Lazarek–Black [25] was used as the corresponding reference for the saturated branch. It was evaluated on the frozen saturated dataset using the same source-file test partitions as the RF analysis. Because the correlation benchmark and RF analysis were not based on identical point-level samples, the comparison is interpreted at the common source-file test-group level.

3. Results

3.1. Database Composition and QC Retention

The source-file-level summary provides a consistent view of operational-branch composition and QC retention without reopening the full merged Parquet. Figure 8 presents the mean within-file shares of operational subcooled, operational saturated, and ambiguous records for each working fluid, with each source file contributing one source-level observation. Figure 9 then compares the QC-retained share across source files for the six fluids.

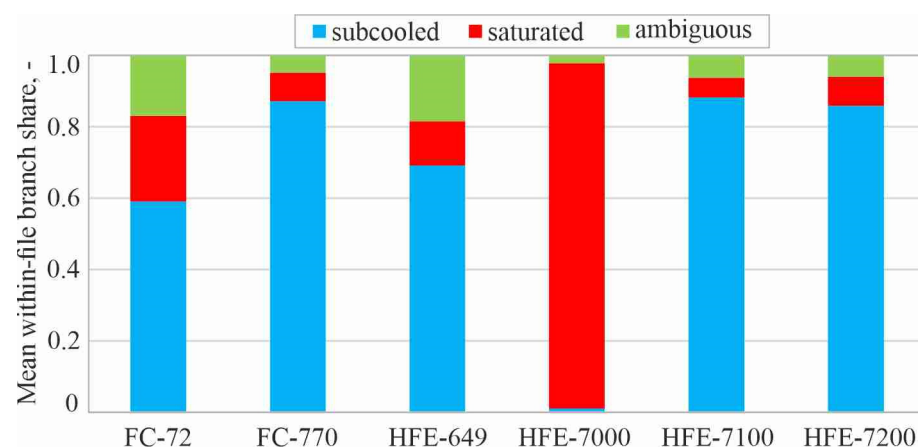


Figure 8. Mean within-source-file operational-branch composition by working fluid. Each source file contributes one source-level set of branch shares.

Figure 8 reveals a pronounced fluid dependence in operational-branch composition. FC-770, HFE-7100, and HFE-7200 are dominated by operational-subcooled records at the source-file level, whereas HFE-7000 is represented by only three source files and is overwhelmingly assigned to the operational saturated branch. FC-72 and HFE-649 contain larger ambiguous and operational-saturated fractions than the other broadly represented fluids. This unequal coverage is a structural property of the archive and must be considered in pooled statistics and branch-resolved modelling.

The distribution of the QC-retained share across source files, grouped by working fluid, is illustrated in Figure 9.

Figure 9 shows that QC retention varies across source files and fluids. The median retained share is highest for FC-770, HFE-7000, HFE-7100, and HFE-7200, while FC-72 and HFE-649 show broader distributions and lower central retention. One HFE-7200 source file has a zero retained share and therefore requires source-file-level review, but it is not silently removed from the archive. The mean source-file QC_{keep} share over all 449 files is approximately 0.860.

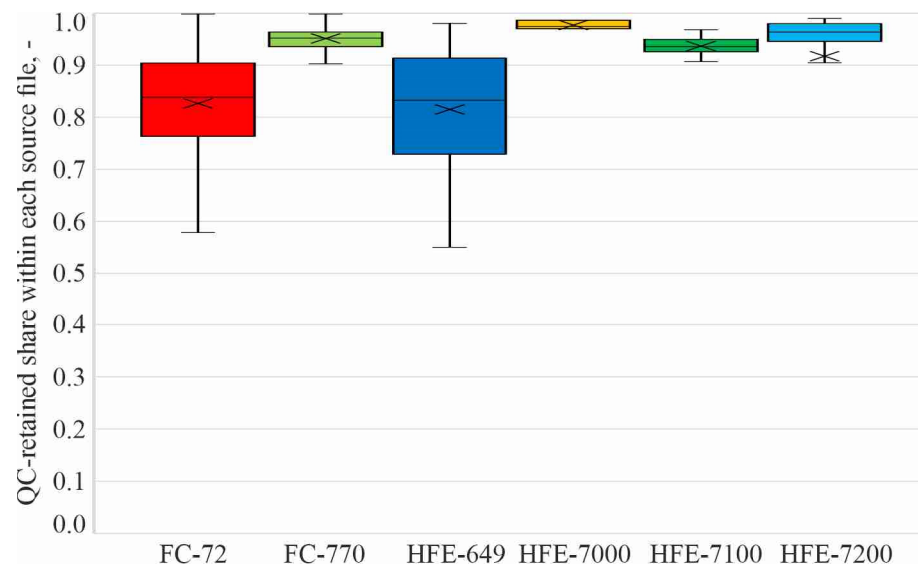


Figure 9. Distribution of the QC-retained share across source files, grouped by working fluid.

3.2. Global Database Characteristics

3.2.1. Distribution of Source-File-Level Heat-Transfer Outputs

The source-file-level summary also permits a robust view of the joint distribution of the two main heat-transfer outputs without allowing very large source files to dominate the display. Figure 10 plots the median α against the median Nu for each source file on logarithmic axes and identifies the dominant operational branch from the largest within-file branch share.

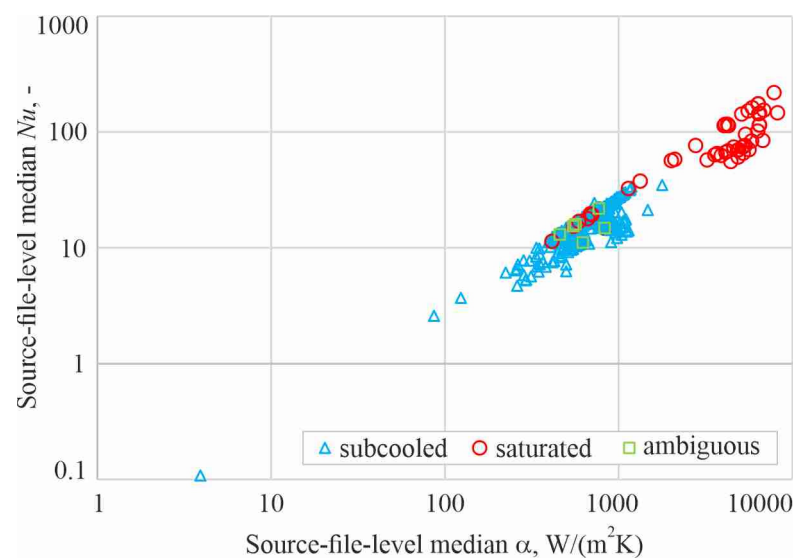


Figure 10. Joint distribution of source-file-level median heat-transfer coefficient and Nusselt number, classified by the dominant within-file operational branch.

Figure 10 shows a strong positive association between source-file-level median α and median Nu , as expected from their related definitions, while also revealing a broad spread across source files. Among the 448 source files with positive paired median values suitable for logarithmic plotting, 394 are operational-subcooled-dominant, 48 are operational-saturated-dominant, and six are ambiguous-dominant. One source file has zero median values of both α and Nu and is therefore omitted from the logarithmic display but retained in the database and in the descriptive summary.

Table 4 provides the numerical companion to Figure 10 using the same source-file-level summary. Each source file contributes one median α and one median Nu value, so the statistics describe between-file variability rather than the much larger number of nested local points. They must not be interpreted as estimates from an equivalent number of independent experiments.

Table 4. Descriptive statistics of source-file-level median heat-transfer coefficient and Nusselt number.

Statistic	Median α , $\text{W m}^{-2} \text{K}^{-1}$	Median Nu , –
Count	449	449
Mean	1092	25.0
Median	726	18.3
P95	4905	75.4
P99	6479	153.3
Maximum	8218	218.6

Table 4 confirms that the source-file-level distributions are right-skewed. For α , the mean exceeds the median by approximately 50%, and the 95th percentile is nearly seven times the median. Nu shows a similar but less extreme pattern. These results support the use of medians and percentile summaries for database characterisation and reinforce the decision not to interpret global arithmetic means as direct surface or fluid effects.

3.2.2. Point-Level Scale Versus Source-File-Level Evidence

The very large point-level database can create a false impression of equally large statistical independence. Figure 11 therefore contrasts the 449 experimental source-file groups with the 64,385,791 local point-level records in the final integrated database. The logarithmic scale is used because the two levels differ by more than five orders of magnitude.

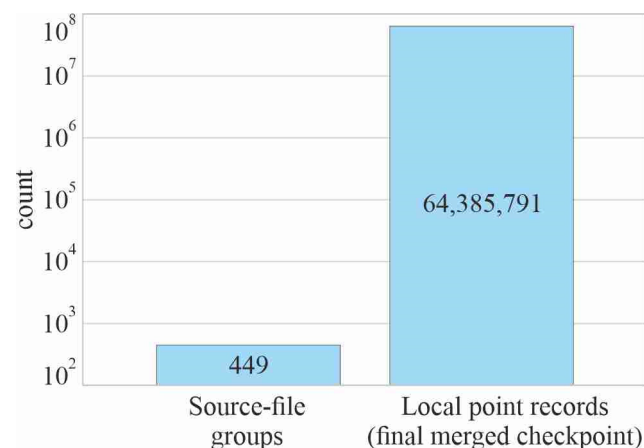


Figure 11. Hierarchical scale of the integrated database: source-file provenance groups versus local point-level records in the final integrated database.

Figure 11 demonstrates the hierarchical scale of the archive: tens of millions of local rows are nested within only 449 provenance groups. The point-level sample size therefore cannot be interpreted as an equivalent number of independent experiments. This scale mismatch is the principal reason for using source file as the grouping variable in validation rather than randomly splitting individual rows.

3.2.3. Coverage Across Working Fluids and Orientations

The integrated archive also has an uneven factorial structure. Figure 12 maps the number of source files for every observed combination of working fluid and nominal orientation angle. The exact nominal orientation angles were retained rather than collapsed into broad classes so that sparse inclined configurations remain visible.

	Nominal orientation angle, °													
	0	15	30	45	60	75	90	105	120	135	150	165	180	270
FC-72	41	1	1	1	2	4	64	1	1	1	1	1	41	47
FC-770	17	1	1	1	1	1	25	1	1	1	1	1	14	10
HFE-649	29	0	0	0	0	0	40	0	0	0	0	0	15	14
HFE-7000	0	0	0	0	0	0	3	0	0	0	0	0	0	0
HFE-7100	7	0	0	0	0	0	9	0	0	0	0	0	3	3
HFE-7200	7	0	0	0	0	0	24	0	0	0	0	0	6	6

Figure 12. Source-file coverage across working fluids and nominal orientation angles. The colour intensity represents the number of source files for each fluid-orientation combination, with darker blue indicating a higher number of files and white indicating zero.

Figure 12 shows that the archive is concentrated at a limited set of principal orientations, particularly 0, 90, 180, and 270 degrees, while intermediate inclined angles are represented by only a few source files. Coverage also differs strongly between fluids. The map is therefore a coverage diagnostic, not a performance comparison, and the pooled data are not interpreted here as balanced evidence of orientation or fluid effects.

3.3. Source-File-Grouped Pilot Random Forest Results

The following four operational-branch–target combinations were evaluated for both the nine-predictor and strict Random Forest predictor sets: subcooled Nu , subcooled α , saturated Nu , and saturated α . Table 5 reports mean $\pm$ standard deviation over the five source-file-grouped splits and includes the two supporting literature references for Nu . For α , RMSE and MAE are expressed in $W\ m^{-2}\ K^{-1}$; for Nu they are dimensionless.

Across the five source-file-grouped splits, the nine-predictor RF achieved $R^2 = 0.887 \pm 0.015$, $RMSE = 2.555 \pm 0.177$, and $MAE = 1.792 \pm 0.048$ for subcooled Nu ; the corresponding subcooled α values were $R^2 = 0.810 \pm 0.043$, $RMSE = 104.1 \pm 14.5\ W\ m^{-2}\ K^{-1}$, and $MAE = 68.1 \pm 2.2\ W\ m^{-2}\ K^{-1}$. Saturated performance was lower and more split-sensitive: for Nu , $R^2 = 0.340 \pm 0.134$, $RMSE = 191.1 \pm 14.9$, and $MAE = 136.7 \pm 8.8$; for α , $R^2 = 0.317 \pm 0.091$, $RMSE = 7337.9 \pm 363.2\ W\ m^{-2}\ K^{-1}$, and $MAE = 5361.6 \pm 260.0\ W\ m^{-2}\ K^{-1}$.

A strict predictor set limited to Re , Pr , L_{obs} , and θ was evaluated on the same five source-file-grouped splits. Performance decreased substantially relative to the nine-predictor set, particularly for the subcooled branch. The strict baseline achieved $R^2 = 0.396 \pm 0.079$ for subcooled Nu , 0.265 ± 0.122 for subcooled α , 0.119 ± 0.067 for saturated Nu , and 0.096 ± 0.066 for saturated α . This confirms that predictors closely related to the reconstructed thermal state contribute materially to the pilot model performance. The strict baseline nevertheless retained non-zero predictive skill, while saturated performance remained weak under both feature sets.

Table 5. Five-split source-file-grouped pilot Random Forest and supporting literature-reference test metrics; splitwise values are reported as mean $\pm$ standard deviation.

Branch	Target	Model/ Reference	Predictors/ Evaluation Basis	R^2 Mean $\pm$ SD	RMSE Mean $\pm$ SD	MAE Mean $\pm$ SD
Subcooled	Nu	Nine-predictor RF	Same fixed 9 predictors; q_w excluded	0.887 ± 0.015	2.555 ± 0.177	1.792 ± 0.048
Subcooled	α	Nine-predictor RF	Same fixed 9 predictors; q_w excluded	0.810 ± 0.043	104.1 ± 14.5	68.1 ± 2.2
Saturated	Nu	Nine-predictor RF	Same fixed 9 predictors; q_w excluded	0.340 ± 0.134	191.1 ± 14.9	136.7 ± 8.8
Saturated	α	Nine-predictor RF	Same fixed 9 predictors; q_w excluded	0.317 ± 0.091	7337.9 ± 363.2	5361.6 ± 260.0
Subcooled	Nu	Strict predictor RF	Re, Pr, L_{obs}, θ	0.396 ± 0.079	5.902 ± 0.263	4.497 ± 0.194
Subcooled	α	Strict predictor RF	Re, Pr, L_{obs}, θ	0.265 ± 0.122	204.8 ± 19.6	146.3 ± 8.8
Saturated	Nu	Strict predictor RF	Re, Pr, L_{obs}, θ	0.119 ± 0.067	221.8 ± 11.1	163.6 ± 6.8
Saturated	α	Strict predictor RF	Re, Pr, L_{obs}, θ	0.096 ± 0.066	8460.9 ± 406.3	6334.5 ± 327.2
Subcooled	Nu	Sieder–Tate [24]	Supporting single-phase reference	-1.244 ± 0.407	11.338 ± 0.605	9.593 ± 0.688
Saturated	Nu	Lazarek–Black [25]	Frozen dataset; same test groups	-0.164 ± 0.095	231.8 ± 20.5	145.0 ± 16.0

For subcooled Nu , Sieder–Tate provided a supporting laminar thermally developing single-phase reference rather than a physical model of the whole subcooled-boiling branch. Over five splits it yielded $R^2 = -1.244 \pm 0.407$, RMSE = 11.338 ± 0.605 , MAE = 9.593 ± 0.688 , and MAPE = $40.3 \pm 3.3\%$; the aggregate MdAPE was 44.4%, with 29.6% and 66.2% of predictions within $\pm 30\%$ and $\pm 50\%$, respectively.

For the saturated branch, the group-aware RF achieved a mean R^2 of 0.340 ± 0.134 over five source-file-grouped splits. As a literature reference, the Lazarek–Black correlation was evaluated independently on the frozen saturated dataset using the same source-file test partitions. It yielded $R^2 = -0.164 \pm 0.095$, RMSE = 231.8 ± 20.5 , MAE = 145.0 ± 16.0 , and MAPE = $83.1 \pm 25.4\%$; the aggregate MdAPE was 49.9%, with 29.0% and 50.2% of predictions within $\pm 30\%$ and $\pm 50\%$, respectively. Although the RF performance remained moderate and split-sensitive, it provided substantially better predictive agreement than this classical minichannel boiling correlation across the heterogeneous multi-fluid dataset. Because the correlation was not evaluated on the identical point-level RF sample, the comparison is interpreted at the common source-file test-group level.

The nine-predictor RF result is summarised visually in Figure 13 using the mean R^2 across the five source-file-grouped splits, with error bars denoting one standard deviation. The figure is a feasibility summary and is not interpreted as a definitive model ranking or as evidence of transfer to unseen fluids, campaigns, surfaces, or configurations.

Figure 13 reinforces the conclusion from Table 5: learnability is strongly dependent on the operational branch. The subcooled tasks provide a proof of concept for the integrated database, whereas the saturated results remain moderate. This difference identifies the branch in which additional physical descriptors and more stringent validation are most needed, without establishing a universal saturated model.

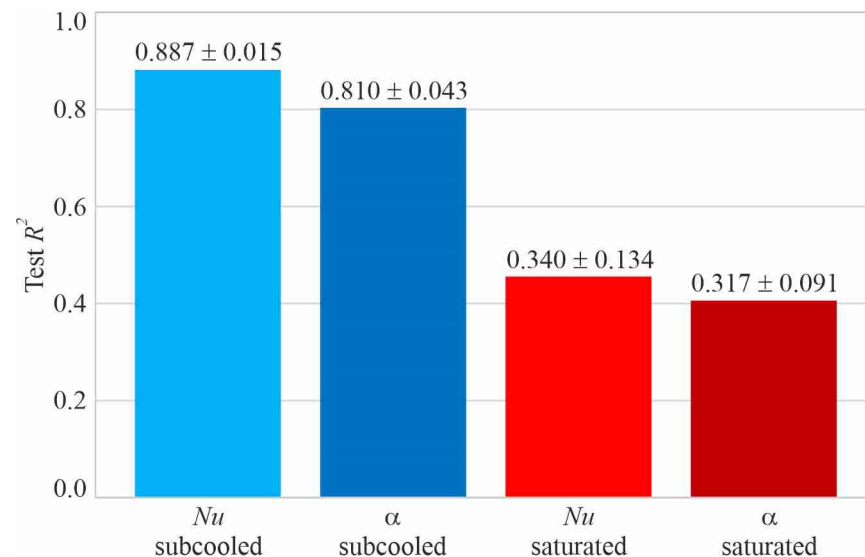


Figure 13. Mean pilot Random Forest test-set R^2 over five source-file-grouped splits; error bars denote one standard deviation. The colours are used to distinguish the operational branches: blue represents the subcooled branch, and red represents the saturated branch; the two shades of blue and red distinguish the Nusselt number and heat transfer coefficient results within each branch.

4. Discussion

4.1. Interpretation of the Pilot Benchmark

The pilot results should not be overinterpreted. Heat-transfer coefficients and Nusselt numbers are derived quantities, and some candidate predictors are mathematically close to their definitions. Source-file-grouped validation controls direct record-level leakage by ensuring that records from the same source file do not occur in both training and test partitions. It does not eliminate target proximity or establish independence between source files belonging to related experimental campaigns, configurations, or operating sequences. The benchmark is therefore interpreted as an assessment of surrogate learnability within the adopted data-reduction framework, with its predictor variants and grouping level stated explicitly.

The present article is self-contained with respect to the database architecture, harmonisation workflow, operational branch definitions, quality-control logic, and pilot source-file-grouped validation. Broader algorithm comparisons and deliberately configuration-held-out transfer studies require separately designed analyses because the archive is strongly imbalanced across fluids, surfaces, orientations, and module configurations; they are not implied by the present pilot results.

4.2. Scope and Limitations

The integrated database is large but not factorially balanced. Fluids, surfaces, orientations, and module configurations have unequal coverage, and the number of point-level records per source file is not uniform. Consequently, global descriptive statistics characterise the archive but do not identify causal effects or guarantee transfer between configurations. The thermographic input is the central longitudinal line of the externally observed heated plate rather than the full two-dimensional detector field. This representation supports the streamwise reconstruction adopted here, but it cannot quantify transverse non-uniformity caused, for example, by parallel-channel flow maldistribution, non-uniform heating, or orientation-dependent effects. The resulting limitation is acknowledged rather than interpreted as evidence of full two-dimensional thermal uniformity.

The workflow propagates measurement inputs into derived heat-transfer quantities, and the influence of temperature-measurement uncertainty on heat-transfer calculations has been examined previously for related rectangular-minichannel experiments [2]. A complete record-level retrospective uncertainty distribution is not available for all 64,385,791 records in the heterogeneous legacy archive. The present database documentation therefore distinguishes available source-level uncertainty information from variables for which record-level uncertainty has not been materialised, rather than assigning uniform retrospective uncertainty values.

The saturated branch was clearly weaker and more split-sensitive than the subcooled branch. This is consistent with the greater physical heterogeneity of two-phase boiling across an archive combining different fluids, surfaces, orientations, and experimental configurations, and with the intentionally limited predictor set. Positive saturated equilibrium vapour quality was not used as a predictor, was not materialised or released for the present analysis, and was not used for operational branch assignment. The present RF results should therefore be interpreted as a group-aware pilot benchmark rather than as a generally transferable boiling model; they do not establish transfer to an unseen campaign, fluid, surface, or configuration.

5. Conclusions

A unified experimental database was assembled from 449 rectangular-minichannel flow boiling source files covering six working fluids and multiple geometric, surface, and orientation configurations.

The Python-based workflow provides a traceable sequence from template-based experimental data and central-line infrared wall-temperature measurements to local thermodynamic reconstruction, operational-branch-aware α and Nu calculation, QC flags, Parquet outputs, source-file-level summaries, and ML subsets.

The database must be analysed at both point and source-file levels because millions of local records are nested within a much smaller number of experimental provenance groups and the coverage is strongly imbalanced.

System-level QC is treated as explicit, auditable logic rather than irreversible outlier deletion. Detailed classification of thermographic line-profile anomalies is outside the scope of the present database-level quality-control framework.

The five source-file-grouped pilot Random Forest splits demonstrated stable predictive performance for the subcooled targets and substantially weaker, more split-sensitive performance for the operational saturated branch. The strict predictor baseline confirmed that reconstructed thermal-state and target-proximal descriptors contribute materially to the nine-predictor set performance, while the supporting Sieder–Tate and Lazarek–Black evaluations provided conventional literature references for the two operational branches. These results are interpreted as group-aware feasibility evidence rather than as proof of campaign-, configuration-, fluid-, or surface-independent transfer.

The present article therefore establishes a documented database and modelling foundation for controlled physical interpretation and explicitly scoped future validation.

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Data Availability Statement: A representative processed data sample, a data dictionary, and supporting documentation describing the database structure and processing workflow will be made publicly available in the GitHub repository “boiling-heat-transfer-minichannel” at <https://github.com/MagdalenaPiasecka/boiling-heat-transfer-minichannel> (accessed on 16 September 2026) upon publication. The full integrated datasets, source metadata, and analysis assets are retained in the PRACE-LAB data-storage environment at Kielce University of Technology and may be made available by the corresponding author upon reasonable request where permitted, subject to the technical and institutional conditions applicable to the archive. For the purpose of Open Access, the authors have applied a CC BY public copyright licence to the Author Accepted Manuscript (AAM) version arising from this submission.

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Nomenclature

Symbols	Definition, unit
A_{ext}	External area used in the heat-loss calculation, m ²
A_{heated}	Electrically heated area, m ²
D_h	Hydraulic diameter, m
k_f	Thermal conductivity of the working fluid, W m ^{−1} K ^{−1}
k_p	Thermal conductivity of the heated plate, W m ^{−1} K ^{−1}
L	Heated length, m
L_{obs}	Observation length of the thermographically analysed section, m
NaN	Not a Number; undefined numerical value, –
Nu	Nusselt number, –
p	Pressure, Pa
p_{abs}	Local absolute pressure, Pa
p_{atm}	Atmospheric pressure, Pa
p_g	Gauge pressure, Pa
Pr	Prandtl number, –
$q_{conv,ext}$	External natural-convection heat-loss flux, W m ^{−2}
q_{loss}	Heat-loss flux referred to the heated area, W m ^{−2}
$q_{loss,ext}$	External heat-loss flux, W m ^{−2}
q_{net}	Net heat flux transferred to the fluid, W m ^{−2}
$q_{rad,ext}$	External radiative heat-loss flux, W m ^{−2}
q_w	Wall heat flux, W m ^{−2}
R^2	Coefficient of determination, –
Re	Reynolds number, –
T_{amb}	Ambient temperature, K
$T_{f,lin}$	Linearly interpolated local bulk-fluid temperature, K
$T_{f,used}$	Reference fluid temperature selected by the operational branch, K
T_{film}	Film temperature for the external heat-loss model, K
T_{sat}	Saturation temperature, K
T_{wall}	Local heated-wall temperature, K
w	Channel width, m
x	Streamwise coordinate, m

Greek letters

α	Local heat-transfer coefficient, $\text{W m}^{-2} \text{K}^{-1}$
δ_p	Heated-plate thickness, m
ΔT_{cond}	Temperature correction for conduction through the heated plate, K
ΔT_{ws}	Wall-to-saturation temperature difference, K
ε	Surface emissivity, –
θ	Nominal module orientation angle, °
σ	Stefan–Boltzmann constant, $\text{W m}^{-2} \text{K}^{-4}$

Subscripts

<i>abs</i>	Absolute pressure
<i>amb</i>	Ambient condition
<i>atm</i>	Atmospheric condition
<i>cond</i>	Conduction correction
<i>conv</i>	External convection
<i>ext</i>	External surface or external-loss reference
<i>f</i>	Fluid
<i>film</i>	Film temperature
<i>g</i>	Gauge pressure
<i>heated</i>	Electrically heated area
<i>in</i>	Channel inlet
<i>lin</i>	Linearly interpolated value
<i>loss</i>	Heat loss
<i>net</i>	Net value after heat-loss correction
<i>obs</i>	Observation
<i>out</i>	Channel outlet
<i>p</i>	Heated plate
<i>rad</i>	Radiation
<i>sat</i>	Saturation
<i>used</i>	Reference value selected by the operational branch
<i>w</i>	Wall
<i>wall</i>	Heated wall (heated plate)

Abbreviations

ANN	Artificial neural network
MAE	Mean absolute error
MAPE	Mean absolute percentage error
MdAPE	Median absolute percentage error
ML	Machine learning
P95	95th percentile
P99	99th percentile
QC	Quality control
QC_{flags}	Pipe-delimited record-level field containing one or more quality-control flags
QC_{keep}	Boolean record-retention indicator equal to True when no QC_{flags} are present and False otherwise
RF	Random Forest
RMSE	Root-mean-square error
SD	Standard deviation
SHAP	SHapley Additive exPlanations

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